Exam 01 (Preliminary) for ENS161
A.Y. 2026-2027, S1

Name _____
Section _____

Instructions. Fill in the information asked above. Submit this questionnaire along with your answer sheets.

Part I. Write your answers in the space provided after each question. For identification questions, each item required is worth one a point. For true-or-false questions, write *only* T or F. For fill-in-the-blank questions, each blank is worth a point.

1. A static system of forces and/or moments acting on a rigid body is called _____ if the unknown support reactions can be found by using only the equilibrium equations.

2. An object is idealized as a ___ if its dimensions are deemed negligible.

3. *True or False.* If a rigid body is in static equilibrium, then the total moment due to the forces (including support reactions) about a point outside the body is zero.

4. Two cables sustain 12- and 14-lb tensions when loaded with a 1-tonne mass. What would be the cable tensions if the mass is halved?

5. An object at rest may be analyzed as a particle even if its dimensions are not negligible, provided that the forces acting on it are ___.

6. Two systems of forces and moments cause the same net force and net moment on an object. The systems are said to be ___.

7. *True or False.* Two forces having equal magnitudes but opposing directions form a couple.

8. With this theorem, the resultant moment of a system of forces about an axis is the algebraic sum of the moments of each of the forces about the same axis.

9. *True or False.* Concurrent forces are not always coplanar.

10. The principle of transmissibility of forces effectively treats a force as a ____ vector.

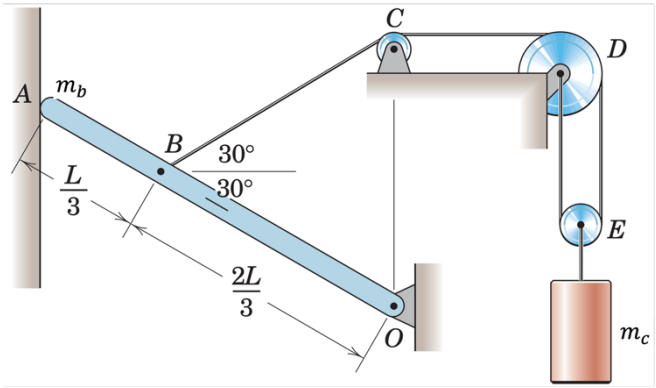
11. *True or False.* A statically determinate system is always stable.

12. *True or False.* The moment of a force $\boldsymbol{F}$ about a point A is perpendicular to both $\boldsymbol{F}$ and the position vector from A to any point in the line of action of $\boldsymbol{F}$.

13. A force vector is characterized by its ___, ___, and ___.

14. Identify three strategies to maximize the moment of a force $\boldsymbol{F}$ about a point A.

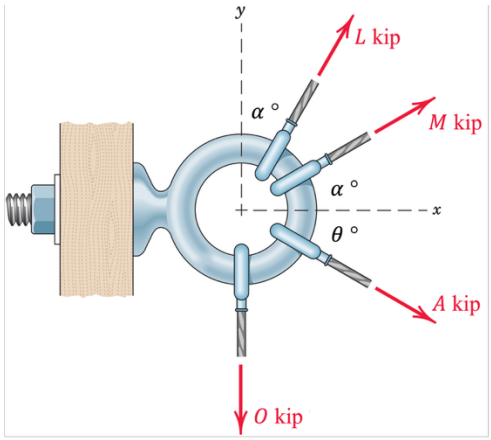
15. Identify three scenarios for which a force does not cause a rotational tendency about an axis.



16. Identify two scenarios wherein the bar is a three-force member. m_b and m_c are the respective masses of the bar and of the cylinder.

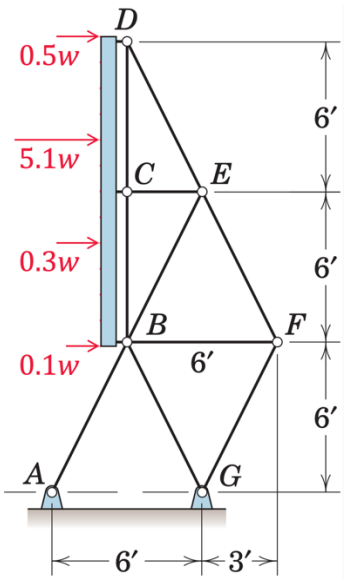
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Part II. Write your solutions and answers on your answer sheets. Where applicable, provide force diagrams. Report final answers in three (3) decimal places. Indicate final answers by enclosing them in boxes.



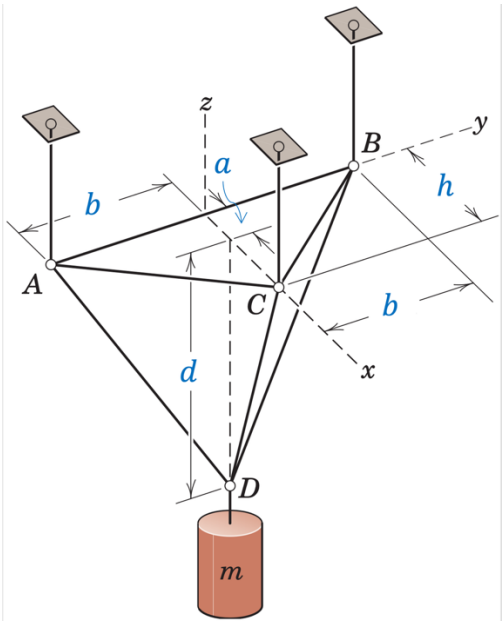
Problem 1. In Alan Parsons' project, an eyebolt in the skyroof system anchors four cables. Suppose that the forces at the perspective of the eyebolt are approximately coplanar, with $\alpha = 30$, $L = 2$, $A = 1.2$, and $O = 1.8$. The four forces must have a purely horizontal resultant. The eyebolt-to-wood joint can sustain no more than 3 kips.

a. [8 pts] Sketch the force diagram using the tip-to-tail method of vector addition.
b. [12 pts] Solve for the limiting values of M and θ.



Problem 2. To befriend the will of the wind is to have one less truss issue. Here, an erratic wind load case is modelled as $0.5w$-, $5.1w$-, $0.3w$-, and $0.1w$-lb forces evenly spaced along the height of the signboard. The truss tower is supported by cable AB and a pin at G. Each of the nine segments of the tower weighs 2 lb/ft, while the signboard weighs 3 lb/ft. Assume that the combined center of gravity of the tower and the signboard is at B.

a. [6 pts] Replace the wind loads with a single-force equivalent.
b. [3 pts] Sketch the FBD of the tower and the signboard. In place of the four wind loads, use the result in Item a.
c. [6 pts] Express the tension in cable AB in terms of w by balancing moments about G.
d. [6 pts] If it is unsafe for cable AB to carry tensions more than 67 lb, what is the maximum allowable value of w?
e. [9 pts] Solve for the reaction at G corresponding to the maximum allowable value of w.



Problem 3. An m-kg model load is hangin' by a thread but is supported by cables AD, CD, and BD. Set a, b, h, and d to 1, 4, 3, and 6 m, respectively. For every kilogram of the model load, determine:

a. [4.5 pts] the moment $\boldsymbol{M}_C$ about C due to the load,
b. [4.5 pts] the moment $\boldsymbol{M}_{AB}$ about the line through A and B due to the load, and
c. [7.5 pts] the tensions T_{AD}, T_{CD}, and T_{BD} developed in cables AD, CD, and BD, respectively.

Sketch an FBD [1.5 pts] for Item c. If cables AD, CD, and BD have respective strengths of 76, 67, and 76 N, solve for:

d. [6 pts] the maximum allowable model load mass, and
e. [6 pts] the corresponding values of T_{AD}, T_{CD}, and T_{BD}.